
EMCH 367 / ELCT 331

Modern Control Systems for Physical Engineering Applications

University of South Carolina

Fall 2026

Cross-listed: EMCH 367 / ELCT 331

Meeting Time: Tue/Thu 10:05–11:20 AM

Location: TBD

Instructor: J.C. Vaught

Office: TBD

Office Hours: TBD

Catalog Description

This course develops practical competency in modeling, feedback control, estimation, and implementation for real engineered systems. The course is concept-first in lecture and application-heavy in assignments. Students learn to build control-ready models of physical systems, design and evaluate feedback controllers, implement controllers digitally, handle key nonlinear effects, and work with multivariable systems using state space methods. The course includes two team projects with physical demonstrations and presentations.

Prerequisites

Students are expected to be fluent in:

- Calculus through multivariable calculus
- Differential equations
- Linear algebra

Students must be comfortable writing and debugging code in Python or MATLAB and producing plots and numerical summaries. **The course does not include a formal review of math prerequisites.**

Learning Outcomes

By the end of the course, students will be able to:

1. **Model physical systems:** Translate a physical system into a control-ready model with explicit assumptions and validity ranges, including actuator and sensor nonidealities

2. **Design feedback controllers:** Design baseline feedback controllers and select appropriate control architectures for real systems
3. **Evaluate performance:** Evaluate performance using standard time-domain metrics and interpret stability evidence in both time and frequency perspectives
4. **Implement digitally:** Implement controllers in discrete time and anticipate common sim-to-hardware failure modes
5. **Handle nonlinearities:** Model and reason about nonlinear effects and robustness
6. **Work with multivariable systems:** Represent and reason about multivariable systems in state space and understand the motivation for native multivariable controller designs
7. **Communicate effectively:** Communicate engineering decisions and experimental evidence in written and oral form

Instructional Approach

Lecture emphasizes **concepts, design reasoning, and engineering workflows** rather than step-by-step derivations. Mathematical derivations and deeper proofs are provided through optional typed notes and video resources.

- **Theory homework** requires formal engagement with selected mathematical tools
- **Application homework** emphasizes implementation, validation, and interpretation

Required Resources

- **Laptop:** Capable of running the course programming environment
- **Software:** The course will standardize on a primary environment (Python or MATLAB) and provide a reference environment for grading
- **Capabilities:** Students must be able to run provided starter code, produce required plots, and submit reproducible results
- **Hardware:** Resources and safety requirements for demos are provided by the course and described on the course site

Assessment

This course is not curved. Final grades are determined by the following components:

Component	Weight
Homework (11 sets, lowest dropped)	50%
Projects	40%
Project 1	15%
Project 2	25%
Exams (two in-term, 5% each)	10%

Final exam slot reserved for optional checkout/overflow.

Homework Structure

Eleven homework sets are assigned; the **lowest score is dropped**. The remaining 10 are weighted by rank, with the lowest grade receiving the highest weight:

Rank	1st	2nd	3rd	4th	5th	6th	7th	8th	9th–10th
Weight	10%	8%	6%	6%	4%	4%	3%	3%	3% each

Each homework includes both **theory** and **application** portions:

- Coding is graded on **final output matching spec exactly**
- All submissions must be **reproducible**

Exams

Two in-term exams (5% each) focus on **applied reasoning**:

- Selecting modeling assumptions
- Choosing control structure
- Interpreting behavior
- Specifying engineering steps
- Diagnosing failures

Optional challenge section: Worth up to 10% bonus on that exam.

Bonus Credit

- Available on individual homework and projects through “above and beyond” extensions
- **Optional bonus research projects:**

- 5-point project: approximately $2\times$ difficulty of Project 2
- 10-point project: approximately $3\text{--}4\times$ difficulty of Project 2
- Must request before **2 weeks prior to final exam**

Course Projects

Projects constitute a significant portion of the course grade and provide hands-on experience with control system design, implementation, and validation. Students will complete two projects throughout the semester, progressing from single-input single-output (SISO) systems to more complex multi-input multi-output (MIMO) systems with machine learning integration.

Team Formation

All projects must be completed in teams. **Single-person projects are not permitted** due to insufficient time for individual presentations and the collaborative nature of real-world engineering practice.

- **Team Size:** 2–3 students per team
- **Formation Period:** Students may self-organize into teams during the first lecture
- **Assignment:** After the first lecture, any remaining unassigned students will be placed into teams by the instructor
- **Team Changes:** **No team changes are permitted until after Project 1 is complete.** Requests for changes after Project 1 must be submitted in writing with justification and are subject to instructor approval.

Project 1: SISO Control Implementation

Weight: 15% of Final Grade

Overview

Project 1 focuses on the design and implementation of a controller for a **single-input single-output (SISO)** system. This project establishes foundational skills in control implementation that will be extended in Project 2.

- Systems may be **continuous-time** or **discrete-time**
- Implementation must be on **real hardware** OR a **detailed simulation** with realistic sensor and actuator models (including noise, quantization, delays, and saturation)
- Projects are **open-ended**—students are encouraged to propose their own systems

Example Systems

The following systems are provided as a starting menu. Students may select from this list or propose an alternative system for approval.

Project 2: MIMO with ML-Based Control

Weight: 25% of Final Grade

Table 1: Example SISO Systems for Project 1

System	Input	Output
Inverted pendulum / cart-pole	Cart force or motor voltage	Pendulum angle
Vehicle steering control	Steering angle	Lateral position or heading
DC motor position control	Motor voltage	Shaft position
DC motor speed control	Motor voltage	Shaft angular velocity
Ball and beam	Beam angle	Ball position
Thermal system regulation	Heater power	Temperature
<i>Custom approved system</i>	<i>TBD</i>	<i>TBD</i>

Overview

Project 2 advances to **multi-input multi-output (MIMO)** systems and introduces machine learning-based control methods. This project emphasizes the comparison between classical and modern control approaches.

- Systems should have **multiple inputs and outputs** (preferably many I/O channels)
- **Must be discrete-time implementation**
- **Must include both:**
 1. An ML-based controller (e.g., reinforcement learning, neural network control, adaptive methods)
 2. A classical controller for comparison (e.g., LQR, PID, state feedback)
- Projects are **open-ended**—students are encouraged to continue/expand their Project 1 system or propose a new system

Example Systems

Table 2: Example MIMO Systems for Project 2

System	Inputs	Outputs
Quadrotor attitude/position	Motor speeds (4)	Roll, pitch, yaw, position (6)
Multi-joint robotic arm	Joint torques/voltages	Joint angles, end-effector pose
Multi-zone HVAC system	Damper positions, fan speeds	Zone temperatures, humidity
Autonomous vehicle	Steering, throttle, braking	Position, velocity, heading
<i>Custom approved system</i>	<i>TBD</i>	<i>TBD</i>

Available Resources

The following resources are available to all project teams:

Demonstration Requirements

All projects require demonstration of the implemented control system. The following requirements apply:

Table 3: Available Project Resources

Resource	Details
3D Printers	Available for fabricating custom mechanical parts
Computers	Workstations with full course software environment
Arduino Microcontrollers	Standard issue for all teams
Raspberry Pi	Available upon request with written justification of computational requirements
Sensor/Actuator Kits	Standardized kits provided to each team
<i>Additional Options:</i>	
Personal Equipment	Students may purchase additional equipment at their own expense
Research Lab Equipment	Students may integrate equipment from their research laboratories with advisor approval

1. **Video Demonstration:** A recorded video demonstration is **required** and must be uploaded with the project submission.
2. **In-Classroom Demonstrations:** Live demonstrations during presentations are permitted *only* if the equipment is small and portable. Large or fragile setups should be demonstrated via video only.
3. **Repeatability:** Demonstrations must show **multiple trials** to establish repeatability of performance. A single successful run is insufficient.
4. **Failure Case Analysis:** Demonstrations must include analysis of failure cases:
 - What conditions cause the controller to fail?
 - What are the stability boundaries?
 - How does the system degrade under disturbances or parameter variations?

Deliverables

Both projects require the following deliverables. All items are mandatory unless explicitly stated otherwise.

Table 4: Required Project Deliverables

#	Deliverable	Description
1	Project Proposal Document	Submitted for instructor approval <i>before</i> work begins. Must include system description, objectives, methodology, and timeline.
2	Code Repository	All source code must be submitted. Code must be reproducible with clear instructions for execution.
3	Team Report	Comprehensive technical report covering theory, design, implementation, results, and analysis.
4	Individual Report	Each team member submits a personal reflection on their contributions and key learnings.
5	Presentation Slides	Submitted <i>before</i> presentation day for review.
6	Live Presentation	In-class presentation followed by Q&A session with instructor and peers.
7	Demo Video	Recorded demonstration with narration explaining the system operation and results.

Grading Criteria

Projects will be evaluated according to the criteria outlined in Table 5. Note that Project 2 places additional emphasis on the rigor of comparison between classical and ML-based controllers.

Important: Project proposals must be approved before work begins.
 Late submissions will be penalized according to the course late policy.
 Academic integrity violations will result in a zero for the project and referral to the Office of Academic Integrity.

Table 5: Project Grading Criteria

Criterion	Evaluation Focus
Correctness of Implementation	Does the controller achieve the stated objectives? Is the implementation technically sound?
Quality of Validation & Testing	Are results validated against theoretical predictions? Is testing comprehensive and systematic?
Robustness of Performance	How does the system perform under disturbances, noise, and parameter variations?
Clarity of Evidence	Are claims supported by clear data and analysis? Are figures and tables well-constructed?
Technical Communication	Quality of written reports, oral presentation, and ability to answer questions.
Comparison (Project 2)	Rigor Depth and fairness of comparison between classical and ML-based controllers. Both approaches must be given equal effort and evaluation.

Course Policies

AI Tools Policy

Context	Policy
Application Assignments	AI tools allowed . Must explain approach, interpret results, and verify correctness.
Theory Assignments	Submissions must reflect the student's own reasoning .
Exams	No AI tools permitted.
Disclosure	Must disclose AI use in a tool-use note with each submission.

Collaboration

- Conceptual discussion with classmates is **allowed and encouraged**
- All submissions must be your **own work**
- Project work must reflect **equitable contribution** from all team members
- External resources must be **cited**

Late Work

- **Homework:** Late submissions generally not accepted (one lowest score dropped)
- **Projects:** Deadlines are firm due to hardware scheduling and presentation logistics

Accessibility

Students with accommodations should coordinate with the instructor **early in the semester** to ensure appropriate arrangements can be made.

Course Schedule

The course meets Tuesdays and Thursdays throughout Fall 2026. The schedule below maps all 28 meetings to specific dates, accounting for university holidays and breaks.

Legend:

- Exam sessions
- Project presentation sessions
- **Bold garnet text** indicates homework due dates and project milestones

Week	Date	Mtg	Type	Topic
1	Thu 8/20	1	Content	Modeling Physical Systems for Control
2	Tue 8/25	2	Content	Modeling Physical Actuators for Control
	Thu 8/27	3	Content	Modeling Sensor Error for Control
3	Tue 9/1	4	Content	Designing Control Systems
	Thu 9/3	5	Content	Measuring Controller Performance HW1 Due: Modeling
4	Tue 9/8	6	Content	Stability Analysis in Time and Frequency Domain
	Thu 9/10	7	Content	Control Architecture HW2 Due: Basic Feedback Design
5	Tue 9/15	8	Content	Introducing State Space Representation
	Thu 9/17	9	Buffer	Integration Studio and Exam 1 Prep HW3 Due: Stability and Architecture
6	Tue 9/22	10	Exam	Exam 1: Modeling, Feedback, Performance, Stability
	Thu 9/24	11	Content	Implementing Control in Discrete Domain Project 1 Proposal Due
7	Tue 9/29	12	Content	Nonlinear System Accounting and Control Robustness
	Thu 10/1	13	Content	Estimating State in Nonlinear Systems HW4 Due: Discrete Implementation
8	Tue 10/6	14	Presentation	Project 1 Presentations Day 1
	Thu 10/8	15	Presentation	Project 1 Presentations Day 2 Project 1 Report Due
9	Tue 10/13	16	Content	Modeling MIMO Systems in State Space
	Thu 10/15	—	<i>Fall Break</i>	<i>No Class</i>
10	Tue 10/20	17	Content	Designing Control Systems for MIMO HW5 Due: Nonlinear and Estimation

Continued on next page

Week	Date	Mtg	Type	Topic
11	Thu 10/22	18	Content	Native MIMO Controllers: LQR, LQG, and MPC
	Tue 10/27	19	Buffer	Integration Studio and Exam 2 Prep HW6 Due: MIMO Systems
	Thu 10/29	20	Exam	Exam 2: Design, Robustness, Implementation
12	Tue 11/3	21	Content	CNN and Transformer Encoders for State Estimation Final Project Proposal Due
	Thu 11/5	22	Content	Transformer World Models for Dynamics and Prediction
13	Tue 11/10	23	Content	Hybrid Model-Based and Learned Control HW7 Due: Learned Perception
	Thu 11/12	24	Content	Differentiable Predictive Control
14	Tue 11/17	25	Content	Diffusion Policies for Trajectory Tracking HW8 Due: Learned Dynamics
	Thu 11/19	26	Content	Generalization and Sim-to-Real for Learning-Based Control
15	Tue 11/24	—	<i>Thanksgiving Break — No Class</i>	
	Thu 11/26	—	<i>Thanksgiving Break — No Class</i>	
16	Tue 12/1	27	Presentation	Final Project Presentations Day 1 HW9 Due: Hybrid and Differentiable Control
	Thu 12/3	28	Presentation	Final Project Presentations Day 2 Final Project Report Due

Assignment Summary

Important Dates

- **First Day of Class:** Thursday, August 20, 2026
- **Labor Day** (University Closed): Monday, September 7, 2026
- **Exam 1:** Tuesday, September 22, 2026
- **Project 1 Presentations:** October 6–8, 2026
- **Fall Break:** Thursday–Friday, October 15–16, 2026
- **Exam 2:** Thursday, October 29, 2026
- **Thanksgiving Break:** Thursday–Friday, November 26–27, 2026
- **Final Project Presentations:** December 1–3, 2026
- **Last Day of Class:** Thursday, December 3, 2026
- **Finals Week:** December 7–11, 2026

Assignment	Due Date	Topic Coverage
HW1	Thu 9/3	Modeling (Meetings 1–3)
HW2	Thu 9/10	Basic Feedback Design (Meetings 4–5)
HW3	Thu 9/17	Stability and Architecture (Meetings 6–8)
HW4	Thu 10/1	Discrete Implementation (Meetings 11–12)
HW5	Tue 10/20	Nonlinear Systems and Estimation (Meeting 13)
HW6	Tue 10/27	MIMO Systems (Meetings 16–18)
HW7	Tue 11/10	Learned Perception (Meetings 21–22)
HW8	Tue 11/17	Learned Dynamics (Meetings 23–24)
HW9	Tue 12/1	Hybrid and Differentiable Control (Meetings 25–26)
Exam 1	Tue 9/22	Modeling, Feedback, Performance, Stability
Exam 2	Thu 10/29	Design, Robustness, Implementation
Project 1 Proposal	Thu 9/24	—
Project 1 Report	Thu 10/8	—
Final Project Proposal	Tue 11/3	—
Final Project Report	Thu 12/3	—

Table 7: Complete assignment schedule with topic coverage.

This syllabus is subject to change. Students will be notified of any modifications.